

N.1

Dimostrare che $\forall n \geq 3 \quad 1 < \frac{F_{n+1}}{F_n} < 2$

$$\underbrace{F_n < F_{n+1}}_{\text{LHS}} < \underbrace{2F_n}_{\text{RHS}}$$

LHS

$$F_{n+1} < F_{n+2}$$

$$F_n + F_{n-1} < F_{n+1} + F_n$$

$$F_{n-1} < F_n + F_{n-1} \quad \text{OK}$$

RHS

$$F_{n+2} < 2F_{n+1}$$

$$F_{n+1} + F_n < 2(F_n + F_{n-1})$$

$$F_{n+1} + F_n < 2F_n + 2F_{n-1}$$

$$F_n + F_{n-1} < F_n + 2F_{n-1}$$

$$F_n < F_n + F_{n-1}$$

$$F_n < F_{n+1} \quad \text{OK}$$

N.2

1) $x \rightarrow (y \rightarrow x)$

2) $[x \rightarrow (y \rightarrow z)] \rightarrow [(x \rightarrow y) \rightarrow (x \rightarrow z)]$

$$\{p \rightarrow q, q \rightarrow r\} \vdash p \rightarrow r$$

$$[p \rightarrow (q \rightarrow r)] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$$

$$(q \rightarrow r) \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$$

MP. $(q \rightarrow r) \rightarrow (p \rightarrow r)$

$$p \rightarrow r$$

N.3

$$\exists x \forall y P(x, y) \wedge \exists x \forall y Q(x, y) \rightarrow \exists x \forall y [P(x, y) \wedge Q(x, y)]$$

$\neg F$

$$\exists x \forall y P(x, y) \wedge \exists x \forall y Q(x, y)$$

$$\neg (\exists x \forall y [P(x, y) \wedge Q(x, y)])$$

$$\exists x \forall y P(x, y)$$

$$\exists x \forall y Q(x, y)$$

$$\neg \exists x \neg \forall y [P(x, y) \wedge Q(x, y)]$$

1) tabeleaux non si chiude

D: Q

$$P(x, y) = x \cdot y = 1$$

$$Q(x, y) = x + y = y$$

$$P(a, b)$$

a fissata

$$Q(c, d)$$

c fissata

$$\neg P(a, e)$$

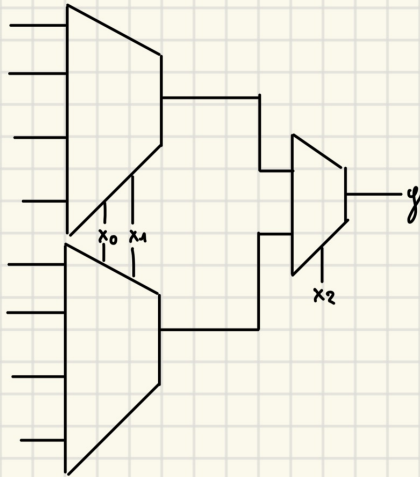
e fissata

$$\neg Q(c, f)$$

f fissata

$$\begin{array}{c}
 \exists x \forall y [P(x,y) \wedge Q(x,y)] \longrightarrow \exists x \forall y P(x,y) \wedge \exists x \forall y Q(x,y) \\
 \hline
 \exists x \forall y [P(x,y) \wedge Q(x,y)] \\
 \neg (\exists x \forall y P(x,y) \wedge \exists x \forall y Q(x,y)) \\
 \hline
 \begin{array}{cc}
 P(a,b) & a \text{ fiktiv} \\
 Q(c,b) & c \text{ fiktiv}
 \end{array} \\
 \swarrow \quad \searrow \\
 \neg (\exists x \forall y P(x,y)) \quad \neg (\exists x \forall y Q(x,y)) \\
 \hline
 \begin{array}{cc}
 P(a,b) & Q(a,b) \\
 \hline
 \times & \times
 \end{array}
 \end{array}$$

N. 9



N. 5

$$a = -15,5$$

$$b = 41DE0000$$

$$0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 0 \ 1 \ 1 \ 1 \ 1 \ 0$$

$$a+b = -15,5 + 27,75 = 12,25$$

$$2^7 + 2^1 + 2^0 = 131$$

$$1 \cdot 101111 \cdot 2^5$$

$$131 - 127 = 4$$

$$1 + 2^{-1} + 2^{-3} + 2^{-4} + 2^{-5} + 2^{-6} \cdot 16$$

$$\left(1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64}\right) 16$$

$$12,25$$

$$16 + 8 + 2 + 1 + \frac{1}{2} + \frac{1}{4} = 27,75$$

$$12 = 1100$$

$$1100.01 \Rightarrow 1,10001 \cdot 10^3$$

$$0,25 \mid 0$$

$$127 + 3 = 130 = 2^7 + 2^1$$

$$0,5 \mid 1$$

$$\begin{array}{ccccccc}
 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0
 \end{array}$$

$$4$$

$$1$$

$$4$$

$$4$$